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City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

KAR Project No. : 713302

Date Reported : 09/26/17

Date Activated : 09/12/17

Date Due : 09/26/17

Date Validated : 09/26/17

Attn : Ms. Cindy Tomaszewski

Project

Description : Analysis of one sludge sample for Fall Biosolids

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. The "Comments" column may contain federal drinking water information including Maximum Contaminant Level set by USEPA, denoted as "MCL*=" and should not be exceeded. Values enclosed in brackets ([]) are USEPA Secondary MCL's which are non-enforceable guidelines for aesthetic quality. Values enclosed in braces ({}) are limits set by Canada, states, or World Health Organization.

If you wish to contact us about this work please mention KAR Project No. . To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink that reads 'David R. Alkema'.

David R. Alkema
Laboratory Manager

KAR Laboratories, Inc. maintains Full Certification status for Bacteriology, Inorganics, Regulated Organics and Synthetic Organics through USEPA, Michigan Department of Public Health and Indiana State Department of Health. This report may only be reproduced in full and not without the written consent of **City of Cadillac, Water Resources**.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **713302**

Date Reported : **09/26/17**

Project

Description: *Analysis of one sludge sample for Fall Biosolids*

Sample ID : <u>"Digester #4"</u>						
Sampled By : DS of City of Cadillac, Works & Resources				Date Received : 09/12/17		
Sample Date : 09/11/17				Sample Type : sludge		
Sample Time : 1315				KAR Sample No. : 713302-01		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep. Hg	Completed		EPA 7471A	09/18/17	MID	
Prep. metals	Completed		EPA 200.7/8	09/13/17	MID	
Aluminum, total	3940	mg/kg total solids	EPA 6010B	09/13/17	JHB	Detection limit = 300
Arsenic, total	22.0	mg/kg total solids	EPA 200.8	09/14/17	NHM	Detection limit = 3.0
Cadmium, total	<3.8	mg/kg total solids	EPA 200.8	09/14/17	NHM	
Calcium, total	28,800	mg/kg total solids	EPA 6010B	09/13/17	JHB	Detection limit = 300
Chromium, total	89.5	mg/kg total solids	EPA 200.8	09/14/17	NHM	Detection limit = 3.0
Copper, total	825	mg/kg total solids	EPA 200.8	09/14/17	NHM	Detection limit = 31
Lead, total, by ICP	<38	mg/kg total solids	EPA 200.8	09/14/17	NHM	
Magnesium, total	4900	mg/kg total solids	EPA 6010B	09/13/17	JHB	Detection limit = 300
Mercury, total	0.57	mg/kg total solids	EPA 7471A	09/18/17	MID	Detection limit = 0.22
Molybdenum, total	24.3	mg/kg total solids	EPA 200.8	09/14/17	NHM	Detection limit = 3.0
Nickel, total	26.6	mg/kg total solids	EPA 200.8	09/14/17	NHM	Detection limit = 3.0
Potassium, total	3480	mg/kg total solids	EPA 6010B	09/13/17	JHB	Detection limit = 300
Selenium, total	5.4	mg/kg total solids	EPA 200.8	09/14/17	NHM	Detection limit = 3.0
Silver, total	<3.8	mg/kg total solids	EPA 200.8	09/15/17	NHM	
Sodium, total	5580	mg/kg total solids	EPA 6010B	09/13/17	JHB	Detection limit = 3000
Zinc, total	1250	mg/kg total solids	EPA 200.8	09/14/17	NHM	Detection limit = 31
Chloride	6230	mg/kg total solids	EPA 300.0A	09/25/17	ALK	Detection limit = 160
Nitrogen, ammonia	39,900	mg/kg total solids	EPA 350.1	09/19/17	JWW	Detection limit = 1300
Nitrogen, nitrate	<16	mg/kg total solids	EPA 300.0A	09/25/17	ALK	
Nitrogen, total kjeldahl	77,900	mg/kg total solids	EPA 351.2	09/13/17	JWW	Detection limit = 2800
Phosphorus, total (as P)	42,300	mg/kg total solids	SM 4500-P B5,E	09/20/17	EIF	Detection limit = 6100
Sulfate	3500	mg/kg total solids	EPA 300.0A	09/25/17	ALK	Detection limit = 310
Waste pH	7.9	S.U.	EPA 9045C	09/14/17	JWW	
Solids, total	3.32	% by weight	EPA 160.3	09/13/17	LPK	Detection limit = 0.01
Solids, volatile	56.18	% of total solids	EPA 160.4	09/13/17	LPK	Detection limit = 0.01

Attest: 
David R. Alkema, Lab Manager

KAR Laboratories, Inc.

March 10, 2017